

TacticalBERT:

Football as a Natural Language

P15: “Cartographers of the Invisible” research track

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INTRODUCTION and MOTIVATION

IDEA:

- **MATCH** as Text
- **POSSESSION** as Sequence
- **EVENT** as Token



make **abstract**
representations visible

POSSESSION (Sequence)	EVENT TYPE (Token)	LOCATION X	LOCATION Y	PLAYER (Squad)
Poss. 9	Carry	50	20	Messi (Barca)
Poss. 9	Pass	49	26	Messi (Barca)
Poss. 9	Shot	99	47	Suarez (Barca)

Poss. 10	Goal Keeper	14	32	Courtois (Real)
Poss. 10	Carry	14	32	Courtois (Real)
Poss. 10	Pass	15	25	Courtois (Real)

Example of two sequences in the StatsbombAPI format

Methodology: MULTIMODAL INPUT SPACE

What (Semantic)



Where (Geometry)

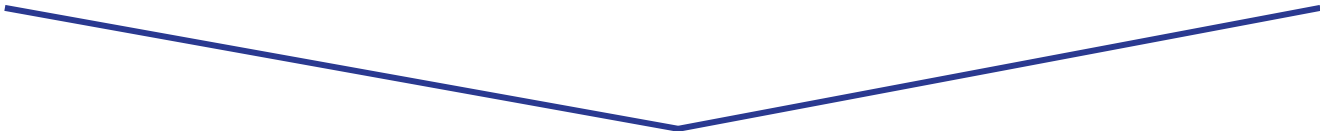


How (Context)

Event type

(x,y) Coordinates

Pressure and Duration



FINAL INPUT
embedding fusion

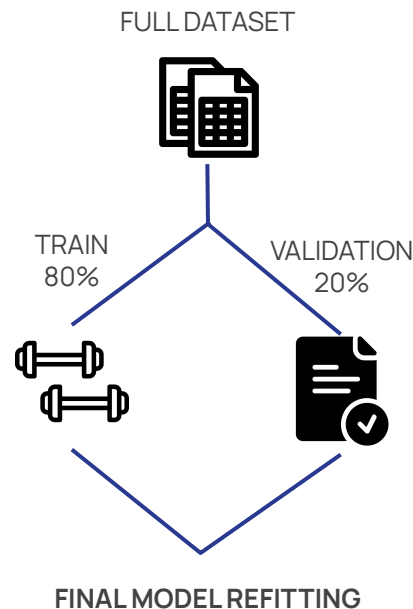
MODEL ARCHITECTURE and TRAINING

Models & Parameters

- BERT
- 4 layers
- 4 heads
- 128-D hidden size

Training

- MLM
- Weighted Loss
- 100 epochs
- Early stopping: **patience** = 15
- +1800 matches and +4M events



PERFORMANCE

- **Stochastic nature** of football :
Interception event
- **Pattern recognition**:
Goalkeeper after *Shot* event
- **Limit of support**: extreme data
scarcity for some events
(*Offside*)

Event type	F1-Score	Support
Pass	0.92	52385
Pressure	0.95	17956
Foul Won	0.98	1564
Interception	0.64	1302
Offside	0.40	13

Overall accuracy : 91%

Macro F1-score : 81%

Best Epoch : 72

PASS CLUSTERS

PASS EMBEDDINGS
from last hidden layer



UMAP
dimensionality reduction



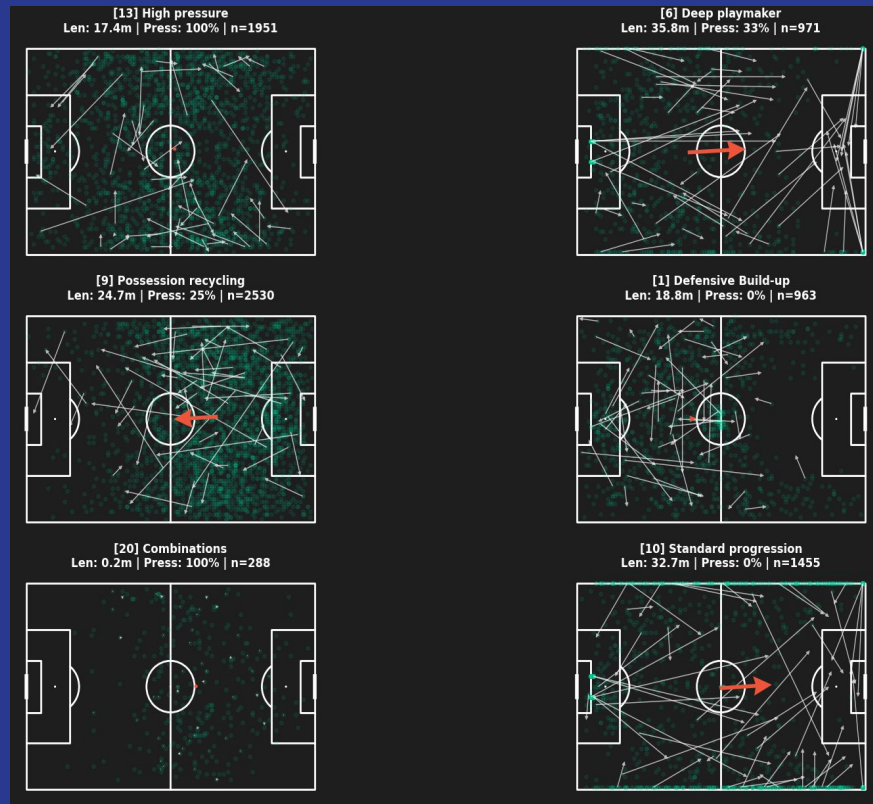
K-MEANS
clustering



DYNAMIC LABELLING
(length, progressive distance, pressure,
coordinates)



PITCH VISUALIZATION
emerging tactical clusters



PLAYER ATLAS

PLAYER EMBEDDINGS

average of all his events embeddings



UMAP

dimensionality reduction



POST-LABELLING

emerging role clustering

- Clear **semantic separation** between roles
- **Overlap** for midfielders-attackers

TacticalBERT Player Atlas



ATTENTION ANALYSIS

- INTERPRETABILITY CHALLENGE



Raw Attention Problem

Residual connections
+
token mixing across layers

- From “*where the model looks*” to “*what the model filters*”
- MULTI-HEAD: functional division of labor
 - **Sequential** heads: focus on chronology of the events
 - **Global** pitch **observers**: elevate spatial spread
 - **Pressure sensors**: focus on pressed events

CRITICAL DISCUSSIONS

- recognition of **distinct tactical phases** and cluster of **similar players**
 - **unpredictable** events and randomness of the sport
 - **bidirectional** nature of the model as a limit
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FUTURE WORKS

- enrich multimodal input space with **new contextual informations**
 - **outliers** analysis in the player atlas
 - explore **co-occurrence bias** related to squad or leagues
 - addressing raw attention problem:
 - **Attention Rollout**
 - **Attention Flow**
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THANKS for the ATTENTION

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